

Spin foam and the origin of the Born rule

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RÉSUMÉ. French translation.

ABSTRACT. The deepest interpretational difficulty of the Everett relative state formulation centers on the question of the origin the Born rule. In this paper, a derivation of the Born rule is proposed that makes use of basic notions in the spin network/spin foam literature. This derivation is proposed to apply to the low energy limit, making use of the original Feynman path integral technique.

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1 Introduction

The MWI is very successful. However, many people still don't like it. A detailed analysis of the MWI by philosophers in the literature lately has identified two important ontologic difficulties. One is easily overcome. The other is not: it is the quantitative problem. It is this problem that this paper is meant to address.

The quantitative problem arises within the MWI at the instant one assumes the Born rule. According to the Born rule, a probability measure $m = |a|^2$ is assigned to each of the N experimental outcomes associated with a measurement, with a being calculated by the Schrodinger equation. The essential difficulty has been stated by Graham: "it is extremely difficult to see what significance such a measure [the Born rule measure] can have when its implications are completely contradicted by

a simple count of the worlds involved, worlds that Everett's own work assures us must all be on the same footing." More recently, Wallace has stated the quantitative problem: why is the probability of a given branch quantified "according to the probability rule (the Born rule), and not (for instance) some other assignment of probabilities to branches?" Greaves picks up on this: "The quantitative problem is perhaps unproblematic in the case of equally weighted superpositions, but has been considered fatal in the case of unequally weighted superpositions . . . it seems that all Everett can say is that each outcome occurs in exactly one branch, which, we might think, will yield equal probabilities if any at all" (emphasis mine).

A detailed discussion of the MWI, or a justification of the benefits of solving the quantitative problem, is beyond the scope of this paper. We will merely take it as given that solving it would be a good thing, so we can move quickly to the model that we pose that eliminates the quantitative problem. Following Greaves, we seek a model that assumes equally weighted superpositions, in which "each outcome occurs in exactly one branch, which . . . will yield equal probabilities."

We will work within the basic framework of Everett's original program (up until he assumes the Born rule), but will also borrow (liberally, and not very rigorously) some basic concepts from the LQG programme [1].

1.1 Terminology

We will begin by borrowing some terminology from LQG. A quantum mechanical spacetime will be represented using a spin foam, which is a (not necessarily connected) graph embedded in a 4-dimensional continuum W ; a spin network is a 3-dimensional hyperslice through a spin foam. The graph is built out of nodes connected by links. Each link is associated with a number j , and each node is associated with a number C . In LQG, the number j corresponds to the area of a surface separating two nodes, and C corresponds to the volume contributed by the node. For this paper, we will interpret j as the "action" of the link, and $S = e \cdot ij$ the "amplitude" of the link. The action is assumed to be directional, so that the action in one direction along a link is the negative of the action in the opposite direction. One of the main results of LQG is that the areas j and volumes C are quantized. We will be exploiting the quantized aspect of spin foams in the proposed model.

1.2 Loop Complex

In this paper, we will be paying special attention to particular configuration which we will call a loop complex. A spin foam may be composed of configurations that are not loop complexes, but we will not be using them for this model.

We define a “loop” as a collection of k nodes and k links, with k an integer at least one, which together form a closed curve, and we will call k the index of the loop. A “unit loop” will be one such that the total action around the loop equals $\pi \hbar$, and in which each link has action $\pi \hbar / k$.

A loop complex of index k is constructed as an ordered succession of unit loops, each with index k . Each node in a unit loop is connected by a link to one of the nodes in the adjacent loop. Any two adjacent nodes on a unit loop must be connected to two adjacent nodes on the adjacent unit loop. A loop complex may be open or closed.

The links that connect two unit loops within a loop complex are given the special designation λ . A loop complex of index k consists of a unit loop, its partner loop, and the k λ 's that connect the unit loops.

An “extended loop” is defined as a single loop within a loop complex that spans two λ . We stipulate that the action around any extended loop is an integer multiple of $2 \pi \hbar$.

From the above, we may deduce that for $k \neq 1$, the sum of the amplitudes of the k λ 's in a loop complex equals zero. Define S_i and S_{i+1} as the actions of two adjacent λ 's in a loop complex. From the Figure, we see that:

$$\pi \hbar / k + S_i + \pi \hbar / k - S_{i+1} = 2 \pi \hbar m.$$

$$S_i - S_{i+1} = 2 \pi \hbar m - 2 \pi \hbar / k$$

Therefore, the actions of two adjacent λ 's are $2 \pi \hbar / k$ out of phase. In other words, the amplitudes of the k λ 's are equidistributed on the unit circle. From the Figure, we arrive at the conclusion that the sum of the amplitudes of the k paths of the complex equals zero. This statement is not true for $k = 1$, in which case there is only a single path λ , whose amplitude is not constrained by the above considerations.

The resolution of the spin foam is arbitrary; higher resolution is achieved by increasing the number of nodes within a given volume of

W. A region that has one loop at one resolution, for example, may have increasingly more loops as the resolution is increased.

1.3 Background manifold in the low energy approximation

Least loops. like a least action principle. Given a configuration of links and associated actions, the density of unit loops is minimized. This leads to

$$\int = \int$$

LQG is background independent. For this paper, however, we will make a low energy approximation so that we can make use of a single flat spacetime manifold M which serves as a background, and a map ϕ for each spin foam which takes the nodes and links in the foam onto points and curves in the background manifold. (A unit loop, we will suppose, is small enough that we may suppose it maps to a point in M .)

Let us suppose that we have a model that tells us that particular configurations of nodes and links in a foam correspond to particular particles or bits of energy. By identifying such configurations in a foam and then applying the map ϕ , we can “translate” a given foam into a distribution of matter (or mass-energy) as a function of location in spacetime M . We will call this an “image” of the spin foam.

Each spin foam is a quantum mechanical spacetime. For a given experimental situation, we will consider a single background manifold M with origin at the observer (see below); but we will require not one, but rather an ensemble of spin foams which exist in superposition to one another. Each foam will have its own unique image in M . These images will agree on some details, and disagree on others, depending on the experimental setup under consideration. In general, the ensemble will be constructed so that each image agrees on the gross (macroscopic) makeup of the experimental apparatus, but will disagree (for example) on the precise details of the states of the individual particles that make up the apparatus.

We will consider a very general quantum mechanical experiment: given a particle that is observed at location s (the source) in M , calculate the relative probability that it will be detected by detector element dn . This can be used to model (among other things) the 2-slit experiment, which Feynman has argued contains the “essential mystery” of QM.

Each foam in the ensemble, therefore, must agree on the following details: a particle emitted from a source s , an array of N detector el-

ements dn , and a system of barriers between the source and detector. Any individual foam will correspond to detection of the particle at a particular dn ; but at the beginning of the experiment, any given pair of foams in the ensemble may or may not agree on where the particle will be detected.

Let us review how the above scenario is analyzed by the Feynman path integral technique. This is performed in the flat background M . The first step is to enumerate the set of “all possible paths” from the source s to the detector dn . Each path has an associated action S and amplitude e^{-iS} . The wavefunction ψ is calculated as the sum of the amplitudes of every path from s to dn .

In the proposed model, each spin foam in the ensemble will have volumes s and dn that map to s and dn in M . Within these volumes, we imagine there to be a large number of unit loops, ranging over all indices; giving rise to a large number of λ ’s stretching between s and dn . Each λ in the foam also maps to a curve in M with endpoints at s and dn . We will stipulate that in our low energy approximation, for each foam in the ensemble, there is a one to one correspondence between λ and ϕ ; furthermore, we stipulate that the action S of λ equals the action S of ϕ as calculated by the FPI.

Consider a small, finite 4-region A in a foam (which may, for instance, correspond to s or dn). Designate the total number of nodes as n and the total number of unit loops as u , with the ratio of loops to nodes defined as $\chi = u/n$. n , u , and χ are local variables characteristic of the 4-region A .

Consider the fact that the number of Feynman paths between s and dn is arbitrarily chosen – we typically consider the “ k most representative” paths, with the understanding that k may be taken to infinity. Note that the number of Feynman paths, as well as the action of each path, is determined at the outset by the experimental setup; these are identical for every foam in the ensemble. The number of paths λ equals the number of nodes, which likewise is the same in every foam. The number of unit loops, however, may vary from one foam to the next. If the indices of every unit loop is $k = 1$, then there is one unit loop per λ , and $\chi = 1$. A different foam in the same ensemble will achieve a lower χ by “coassociating” k λ ’s into a single loop complex. Different foams may have the same χ , but differ with respect to which λ ’s are coassociated with which. The details of coassociation are not purely arbitrary, but are constrained by the actions of the λ ’s

(which do not change from one foam to the next): k lambda's can be coassociated only if their amplitudes are equally spaced on the unit circle. The minimum possible value of χ , therefore, would correspond to a foam in which most of the lambda's belong to loop complexes with high indices, leaving the fewest possible number of lambda's to exist independently (belong to loop complex with index $k = 1$).

We will pay special attention to those foams for which χ takes its minimum value. In these foams, every lambda has a high index, except for a very few which cannot be paired with any others. It may be seen (see the Figure) that the sum of the amplitudes of every path from s to dn equals the number of index $k = 1$ paths from s to dn .

1.4 Particle model

Our simple model of a particle in a spin foam will be to stipulate the following: a particle trajectory corresponds to a closed loop complex of index $k = 1$. In other words, if a particle traverses spacetime regions A and B in a spin foam, then there must exist a closed loop complex of index one, with one unit loop in A and the other in B . This will, in turn, mean that out of all the paths lambda connecting A to B , two of them are "special" in the sense that they correspond to the two ways around the closed loop complex associated with the particle.

1.5 Observer and probability

In accordance with the MWI, any piece of matter – a computer, a person, or even a single lonesome particle – may play the role of the observer. For our purposes, we will define the observer (at the time t_0) as a 3-volume with a 2-sphere as boundary. Given a particular choice for the observer, there will be a corresponding ensemble E of spin foams, which are considered to exist in superposition to one another. The spin foams in E are subject to a single constraint: each spin foam must contain a representation of the observer. In other words, there must exist a 3-volume somewhere in the foam which exactly "looks like" the observer.

Each spin foam in E will have its own image in M . By constraining the foams that are allowed in the ensemble E , the state of the observer constrains the possible images in M . These images may agree on some details, and disagree on others. Imagine, for instance, that each image shows a particle at spacetime location a . We would say that the observer possesses an internal recording of the existence of the particle at

a. As another example, imagine that each image shows a piece of experimental apparatus at some location, but that the microscopic details of the apparatus differ from image to image. This would be sufficient to conclude that the observer possesses an internal recording of some particular experimental setup. In this manner, the observer is said to possess an internal recording of the state of the outside world.

Within a given spin foam, the observer has a unique evolution in time, which is defined by parallel transport forward in time. (The time direction is part of the definition of the observer.) But the time evolution of the state of the observer is not necessarily the same for different spin foams in the ensemble. Suppose that the state of the observer is plotted on a graph, with the y-axis representing the observer state, and the x-axis representing time. Each spin foam gives rise to a single curve; but if we superimpose all the curves from every foam in the ensemble, we get a tree-like diagram. Given an observer in some definite initial state, we can calculate the ensemble E as well as the associated tree diagram that prescribes its possible evolutions. Consider a point on the tree that splits into N branches. What rule do we use to tell us the probability of evolution along any given branch? We could use the Born rule; this is essentially what Everett did. But we seek a solution to the quantitative problem, and so we will suppose the EPP, which states: each branch is realized with probability $m = 1/N$. Our goal for the remainder of this paper will be to demonstrate how the Born rule may emerge as a coarse-grained approximation to a system that is fundamentally governed by the EPP at the fine-grained level. This will require two additional assumptions about the nature of the fine-grained makeup of tree diagrams.

We will assume that the tree diagram of the above experiment, at the microscopic level, takes the two-step configuration shown in the Figure. The first stage corresponds to the observation of the emission of the particle at the source, which occurs at the instant that the event “emission of particle from source” is in the past light cone of the observer. The process of observation includes observation of the variable χ , which is a local variable describing the region s . The first step in the tree diagram minimizes χ . As was discussed above, if χ takes its minimum value, then we have the relation: $a_n = (\chi)$.

Once the first stage of the tree diagram is passed (once the observer has fully registered the emission of the particle at s), we may consider the number of index $k = 1$ paths from s to d_n to be established (frozen,

collapsed). The next step in the tree diagram corresponds to the observation of the particle at one of the detector elements. Recall that out of all the index $k = 1$ lambda's connected to s , two of them are "special" in the sense that they uniquely correspond to the particle. We consider that the process of observation of the particle at dn corresponds to identification of these two paths. There are $\chi(\chi - 1)$ ways to choose two of them as special. According to the model, there is one branch on the tree diagram for each possible pairwise combination. Thus, the number of branches associated with detector element dn is proportional to χ^2 . According to the EPP, the probability of detection at dn is likewise χ^2 . Since we have already established that $\chi = \chi_{an}$, so that $\chi^2 = \chi_{an}^2$, we may conclude that the proposed scheme predicts the same probability as the FPI.

1.6 Discussion

The purpose of this paper has been to propose a variation of the MWI in which probabilities are fundamentally calculated by the EPP rather than the Born rule. Nevertheless, it has been shown that this model makes the same predictions as the FPI technique, and is therefore consistent with experiment. In other words, it has been demonstrated that the Born rule emerges at the coarse-grained level, with probabilities in fact governed by the EPP at the fine-grained level.

The proposed scheme relies on a number of axioms and assumptions, some more easy to justify than others. How can we justify the low energy approximation – in particular, the one-to-one correspondence between lambda and the Feynman paths? Why should the tree diagram have the microstructure that it has?

locality

¹ etc.

References

- [1] Loop and spin foam quantum gravity: a brief guide for beginners.

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¹Footnote